

MIDTERM EXAMINATION

Spring 2010

MTH101- Calculus And Analytical Geometry

<http://groups.google.com/group/vuZs>

Time: 60 min

Marks: 40

Calculus And Analytical Geometry – I

Solved by : Gulshan Ali (Hafizabad)

gulshanvu@yahoo.com

Question No: 1 (Marks: 1) - Please choose one

$30^0 = \underline{\hspace{2cm}}$

$\frac{\pi}{3}$



$\frac{\pi}{4}$



$\frac{\pi}{6}$



$\frac{\pi}{2}$



<http://groups.google.com/group/vuZs>

Question No: 2 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then which of the following statement is correct?

- ▶ f is a decreasing function
- ▶ f is an increasing function
- ▶ f is a constant function

Question No: 3 (Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

- ▶ $\pm \frac{k\pi}{2} (k = 1, 3, 5, \dots)$
- ▶ $\pm \frac{k\pi}{2} (k = 2, 4, 6, \dots)$
- ▶ $\pm \frac{k\pi}{2} (k = 1, 2, 3, 4, 5, 6, \dots)$
- ▶

Question No: 4 (Marks: 1) - Please choose one

$\lim_{x \rightarrow \infty} (-2x) =$

- ▶ -2
- ▶ 0
- ▶ 2
- ▶ Does not exist

<http://groups.google.com/group/vuZs>

Question No: 5 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx} [f][g] =$$

- ▶ $\frac{[f'] [g] - [f] [g']}{g^2}$
- ▶ $[f'] [g']$
- ▶ $[f'] [g] + [f] [g']$
- ▶ $[f'] [g] - [f] [g']$

Question No: 6 (Marks: 1) - Please choose one

The solution set of the inequality $|x+4| \geq 2$ is

- ▶ $(-\infty, -6] \cup [2, +\infty)$
- ▶ None of these
- ▶ $(-\infty, 6] \cup [-2, +\infty)$
- ▶ $(-\infty, -6] \cup [-2, +\infty)$

<http://groups.google.com/group/vuZs>

Question No: 7 (Marks: 1) - Please choose one

A line is called a tangent line to the circle if it meets the circle at precisely

- ▶ One point
- ▶ Two points
- ▶ Infinite points

Question No: 8 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- ▶ f is an increasing function.
- ▶ f is a decreasing function.
- ▶ f is a constant function.

Question No: 9 (Marks: 1) - Please choose one

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

- ▶ $3x^7 + 2$
- ▶ $24x^7 + 2$
- ▶ $3x^9 + 2x^2$

▶ $24x^9 + 2x^2$

Question No: 10 (Marks: 1) - Please choose one

π is anumber

- ▶ Integer
- ▶ Rational
- ▶ **Irrational**
- ▶ Natural

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Question No: 11 (Marks: 1) - Please choose one

The set $\{x: a \leq x \leq b\}$ can be written in the form of interval

- ▶ (a,b)
- ▶ (a,b]
- ▶ **[a,b]**

Question No: 12 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx} \left[\frac{f}{g} \right] =$$

▶ $\frac{[g][f'] - [f][g']}{g^2}$

▶ $\frac{[g'][f] - [f'][g]}{g^2}$

▶ $\frac{[g][f'] - [f][g']}{f^2}$

▶ $\frac{[g'][f] - [f'][g]}{f^2}$

Question No: 13 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about -----axis

- ▶ X-axis
- ▶ Y-axis
- ▶ Origin

Question No: 14 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow -7} \frac{x^2 - 49}{x + 7}$$

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- ▶ -14
- ▶ 0
- ▶ ∞
- ▶ Limit does not exist

Question No: 15 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- ▶ Product
- ▶ Sum
- ▶ Difference
- ▶ Composition

Question No: 16 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow a} f(x) = \dots \text{where } f(x) = k$$

The _____ (k is a constant)

- ▶ k+2
- ▶ k+1
- ▶ k

Question No: 17 (Marks: 1) - Please choose one

For any number $\epsilon > 0$ if we can find an open interval (x_0, x_1) on the x-axis containing the point "a" such that $L - \epsilon < f(x) < L + \epsilon$ for each x in (x_0, x_1) except the possible x = a then we say $\lim_{x \rightarrow a} f(x) =$ _____

- ▶ L

- ▶ $L - \varepsilon$
- ▶ $L + \varepsilon$
- ▶ $L + 1$

Question No: 18 (Marks: 1) - Please choose one

If $2x - y = -3$ then $\frac{dy}{dx} =$

- ▶ 2
- ▶ -2
- ▶ 0
- ▶ -3

Question No: 19 (Marks: 1) - Please choose one

The graph of the equation $y = x^2 - 4x + 5$ will represent

- ▶ Parabola
- ▶ Straight line
- ▶ Two straight lines
- ▶ Ellipse

Question No: 20 (Marks: 1) - Please choose one

The equation of line of the form $y - y_1 = m(x - x_1)$ is known as

- ▶ Point-slope form
- ▶ Two points form
- ▶ Intercepts form
- ▶ Slope intercept form

Question No: 21 (Marks: 2)

If $y = 5 \cos(x^2 + 1)$ Find dy/dx by using "The chain rule".

Question No: 22 (Marks: 2)

Prove that $\lim_{x \rightarrow 1} (x + 5) = 6$, using the definition of limit.

Question No: 23 (Marks: 3)

Find an equation of the tangent line to the curve

$$y = \frac{2}{x^2 + x}$$

at the point where $x = 1$

Question No: 24 (Marks: 3)

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{3x}$$

Compute

Question No: 25 (Marks: 5)

$$f(x) = x^4 - 4x^3 + 4x^2$$

Find all critical points of

Question No: 26 (Marks: 5)

$$\frac{dy}{dx} \quad y = x^2 (\cot x) - \frac{1}{x^2}$$

Find if

Best link for solved papers

<http://groups.google.com/group/vuZs>

MIDTERM EXAMINATION
Spring 2009
MTH101- Calculus And Analytical Geometry (Session - 6)

Time: 60 min

Marks: 40

Calculus & Analytical Geometry-I

Gulshan Ali (Hafizabad)
gulshanvu@yahoo.com

Question No: 1 (Marks: 1) - Please choose one

_____ The
set $\{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is known as set of

- ▶ Natural numbers
- ▶ **Integers**
- ▶ Whole numbers
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

_____ The
$$h(x) = \frac{1}{(x-2)(x-4)}$$

domain of the function is

- ▶ **$(-\infty, 2) \cup (2, 4) \cup (4, +\infty)$**
- ▶ $(-\infty, 2) \cup \{2, 4\} \cup (4, \infty)$
- ▶ $(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$
- ▶ All of these are incorrect

Question No: 3 (Marks: 1) - Please choose one

If $\lim_{x \rightarrow a} f(x) = L$ then the inequality $(L - \varepsilon) < f(x) < L + \varepsilon$ holds in any subset of the interval

- ▶ $(a - \delta, a) \cup (a, a + \delta)$
- ▶ $(a - 1, a) \cup (a, a + 1)$
- ▶ $(a - \varepsilon, a) \cup (a, a + \varepsilon)$
- ▶ None of these

Question No: 4 (Marks: 1) - Please choose one

$L - \varepsilon < f(x) < L + \varepsilon$ Can be written as

- ▶ $|f(x) - L| < \varepsilon$
- ▶ $|f(x) - L| > \varepsilon$
- ▶ $|f(x) - L| < \varepsilon + 1$
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

If a function satisfies the conditions

$f(c)$ is defined

$$\lim_{x \rightarrow c^+} f(x)$$

Exists

$$\lim_{x \rightarrow c^+} f(x) = f(c)$$

Then the function is said to be

- ▶ Continuous at c
- ▶ Continuous from left at c
- ▶ Continuous from right at c

► None of these

Question No: 6 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\sec x] = \text{-----}$$

► $\frac{\sin x}{1 - \sin^2 x}$

► $\frac{-\sin x}{1 - \sin^2 x}$

► $\frac{1}{1 - \sin^2 x}$

► None of these

Question No: 7 (Marks: 1) - Please choose one

$$\log_b ac = \text{-----}$$

► $\log_b a + \log_b c$

► $\log_a b + \log_c b$

► $\log_{a+c} b$

► None of these

Question No: 8 (Marks: 1) - Please choose one

$$\log_b a^r = \text{-----}$$

► $a \log_b r$

► $r \log_b a$

- ▶ $b \log_a r$
- ▶ None of these

Question No: 9 (Marks: 1) - Please choose one

_____ If $f''(x) < 0$ on an open interval (a,b) then f is ----- on (a,b)

- ▶ None of these
- ▶ Concave up
- ▶ Concave down
- ▶ Closed

Question No: 10 (Marks: 1) - Please choose one

_____ If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

- ▶ Minima
- ▶ Maxima
- ▶ None of these

Question No: 11 (Marks: 1) - Please choose one

_____ A line is called a tangent line to the circle if it meets the circle at precisely

- ▶ One point
- ▶ Two points
- ▶ Infinite points
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

_____ The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at and radius

- ▶ $(-4,1), \sqrt{6}$
- ▶ $(-4,1), 6$

- ▶ $(-4, -1), \sqrt{6}$
- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

$\lim_{x \rightarrow a} f(x)$ where $f(x) = k$

(k is a constant) is equal to

- ▶ $k+2$
- ▶ $k+1$
- ▶ **k**
- ▶ kf

Question No: 14 (Marks: 1) - Please choose one

any polynomial $P(x) = c_0 + c_1x + \dots + c_nx^n$ and any real number a
 $\lim_{x \rightarrow a} P(x) = c_0 + c_1a + \dots + c_na^n =$

- ▶ **$P(a)$**
- ▶ $P(a+1)$
- ▶ $P(a-1)$
- ▶ $\frac{1}{P(a)}$
- ▶

Question No: 15 (Marks: 1) - Please choose one

Polynomials are always Function

- ▶ **Continuous**
- ▶ Discontinuous

Question No: 16 (Marks: 1) - Please choose one

$\frac{D}{Dx}[dh(x)] = \text{---}$

where d is a constant

- ▶ $dh(x)$
- ▶ $dh'(x)$
- ▶ 0
- ▶ None of these

Question No: 17 (Marks: 1) - Please choose one

graph $x = y^2$ is symmetric about

The

- ▶ X-axis
- ▶ Y-axis
- ▶ Origin
- ▶ None of these

Question No: 18 (Marks: 1) - Please choose one

Consider two function $f(x) = 3\sqrt{x}$ and $g(x) = \sqrt{x}$ what is true about these functions

- ▶ $f(x).g(x) = 3x$
- ▶ $\frac{f(x)}{g(x)} = 3x$
- ▶ $f(g(x)) = 3x$
- ▶ None of these

Question No: 19 (Marks: 1) - Please choose one

formula $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is called with respect to x of the function f

The

- ▶ Derivative
- ▶ Slope
- ▶ Tangent
- ▶ None of these

Question No: 20 (Marks: 1) - Please choose one

Suppose that f and g are differentiable function of x then $\frac{d}{dx}\left(\frac{f}{g}\right)$

☒ $\frac{g \cdot f' - f \cdot g'}{g^2}$

☐ $\frac{g \cdot f' + f \cdot g'}{g^2}$

☐ $\frac{g \cdot f' - f \cdot g'}{g}$

☐ None of these

Question No: 21 (Marks: 2)

If $\frac{dy}{dx} = -\frac{3yx^2 + 1}{28y^3 + x^3}$ then find the slope of the tangent line at the point (2, 0).

Question No: 22 (Marks: 3)

Let $f(x) = \begin{cases} \frac{x^2 - x - 2}{x + 1} & \text{if } x \neq -1 \\ -3 & \text{if } x = -1 \end{cases}$

At what points the function f is continuous and discontinuous? At point of discontinuity if any explain why it is discontinuous?

Question No: 23 (Marks: 5)

Differentiate w.r.t. x by chain rule $y = \sqrt{x^2 + 1}$

Question No: 24 (Marks: 10)

Evaluate the following limit.

$$\lim_{y \rightarrow -2} g(y) \text{ where, } g(y) = \begin{cases} y^2 + 5 & \text{if } y < -2 \\ 3 - 3y & \text{if } y \geq -2 \end{cases}$$

Haftizabad Campus

MIDTERM EXAMINATION
Fall 2009
MTH101- Calculus And Analytical Geometry

Time: 60 min
Marks: 42

Calculus & Analytical Geometry-I

Gulshan Ali (Hafizabad)
gulshanvu@yahoo.com

Question No: 1 (Marks: 1) - Please choose one

The base of the natural logarithm is

► 2.71

► 10

► 5

► None of these

Question No: 2 (Marks: 1) - Please choose one

A line $x = x_0$ is called ----- for the graph of a function f if
 $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

► Horizontal asymptotes

► None of these

► Vertical asymptotes

Question No: 3 (Marks: 1) - Please choose one

If a function satisfies the conditions

$f(c)$ is defined

$\lim_{x \rightarrow c^+} f(x)$

Exists

$\lim_{x \rightarrow c^+} f(x) = f(c)$

Then the function is said to be

► Continuous at c

► Continuous from left at c

► Continuous from right at c

► None of these

Question No: 4 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a,b) then f is ----- on (a,b)

- ▶ None of these
- ▶ Concave up
- ▶ Concave down
- ▶ Closed

Question No: 5 (Marks: 1) - Please choose one

Suppose that f and g are differentiable function of x then

▶ $\frac{g \cdot f' - f \cdot g'}{g^2}$

▶ $\frac{g \cdot f' + f \cdot g'}{g^2}$

▶ $\frac{g \cdot f' - f \cdot g'}{g}$

- ▶ None of these

Question No: 6 (Marks: 1) - Please choose one

A line $x = x_0$ is called ----- for the graph of a function f if $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

- ▶ Horizontal asymptotes
- ▶ None of these
- ▶ Vertical asymptotes

Question No: 7 (Marks: 1) - Please choose one

If $y = \frac{1}{1-x}$ then $\frac{dy}{dx} =$

▶ $\frac{1}{(1-x)^2}$

▶ $\frac{-1}{(1-x)^2}$

Question No: 8 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx}[f][g] =$$

- ▶ $\frac{[f'] [g] - [f] [g']}{g^2}$
- ▶ $[f'] [g']$
- ▶ $[f'] [g] + [f] [g']$
- ▶ $[f'] [g] - [f] [g']$

Question No: 9 (Marks: 1) - Please choose one

If $x^2 + y^2 = 9$ then $\frac{dy}{dx} =$

- ▶ $\frac{x}{y}$
- ▶ $\frac{-x}{y}$
- ▶ $\frac{-y}{x}$
- ▶ $\frac{y}{x}$

Question No: 10 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

- ▶ Minima
- ▶ Maxima
- ▶ None of these

Question No: 11 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

- ▶ Minima
- ▶ Maxima
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

If

$f''(x) > 0$ on an open interval (a,b), then which of the following statement is correct?

- ▶ f is concave up on (a, b).
- ▶ f is concave down on (a, b).
- ▶ f is linear on (a, b).

Question No: 13 (Marks: 1) - Please choose one

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

- ▶ $\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$
- ▶ $\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$
- ▶ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$
- ▶ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$

Question No: 14 (Marks: 1) - Please choose one

If $x > 0$ then $\frac{d}{dx}[\ln x] =$

- ▶ 1
- ▶ x
- ▶ $\frac{1}{x}$
- ▶ $\ln \frac{1}{x}$

Question No: 15 (Marks: 1) - Please choose one

$\log_b ac =$

- ▶ $\log_b a + \log_b c$
- ▶ $\log_a b + \log_c b$
- ▶ $\log_{a+c} b$
- ▶ None of these

Question No: 16 (Marks: 1) - Please choose one

$$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$$

- ▶ $\log_b c$
- ▶ $1 - \log_b c$
- ▶ $-\log_b c$
- ▶ $1 + \log_b c$

Question No: 17 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

- ▶ $\log_b t$
- ▶ $1 - \log_b t$
- ▶ $1 + \log_b t$
- ▶ $-\log_b t$

Question No: 18 (Marks: 1) - Please choose one

If we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$

- ▶ $\frac{-x}{y}$
- ▶ $\frac{x}{y}$
- ▶ $\frac{-y}{x}$
- ▶ None of these

Question No: 19 (Marks: 1) - Please choose one

$$\log_b a^r = \underline{\hspace{2cm}}$$

▶ $a \log_b r$

▶ $r \log_b a$

▶ $\frac{\log_b a}{\log_b r}$

▶

▶ $\log_b a + \log_b r$

Question No: 20 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

▶ f is an increasing function.

▶ f is a decreasing function.

▶ f is a constant function.

Question No: 21 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

▶ f is an increasing function.

▶ f is a decreasing function.

▶ f is a constant function.

Question No: 22 (Marks: 5)

Differentiate w.r.t. x by chain rule $y = \sqrt{x^2 + 1}$

ASSALAM O ALAIKUM all fellows
ALL IN ONE Mega File
MTH101 Midterm PAPERS,
MCQz & subjective
Created BY Farhan & Ali
BS (cs) 3rd sem
Hackers Group
Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

MIDTERM EXAMINATION
Fall 2009

MTH101- Calculus And Analytical Geometry (Session - 5)

Ref No: 1087754

Time: 60 min

Marks: 42

Student Info	
StudentID:	BC090400057
Center:	OPKST
ExamDate:	12/17/2009 12:00:00 AM

For Teacher's Use Only

Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

► Minima

► **Maxima**

► None of these

Note: Maxima (If Maxima refers to local maximum)

Question No: 2 (Marks: 1) - Please choose one

A line $x = x_0$ is called ----- for the graph of a function f if $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

► Horizontal asymptotes

► None of these

► **Vertical asymptotes**

Question No: 3 (Marks: 1) - Please choose one

A line $y = y_0$ is called a for the graph f if $\lim_{x \rightarrow +\infty} f(x) = y_0$ or $\lim_{x \rightarrow -\infty} f(x) = y_0$

- ▶ Vertical asymptotes
- ▶ Horizontal asymptotes
- ▶ None of these

Question No: 4 (Marks: 1) - Please choose one

According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number, then

$$\frac{d}{dx}[x^n] =$$

- ▶ x^{n-1}
- ▶ nx^{n-1}
- ▶ nx^{n+1}
- ▶ $(n-1)x^{n+1}$

Question No: 5 (Marks: 1) - Please choose one

$$y = \frac{1}{1-x} \quad \frac{dy}{dx} =$$

then

- ▶ 1
- ▶ -1
- ▶ $\frac{1}{(1-x)^2}$
- ▶ $\frac{-1}{(1-x)^2}$

Question No: 6 (Marks: 1) - Please choose one

If

If

$xy = 4$ then $\frac{dy}{dx} =$

► 0

► $\frac{-1}{x^2}$

► $\frac{4}{x^2}$

► $\frac{-4}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

If

$2x - y = -3$ then $\frac{dy}{dx} =$

► 2

► -2

► 0

► -3

Question No: 8 (Marks: 1) - Please choose one

$\frac{d}{dx}[\sec x] =$ _____

► $\frac{1}{1 + \sin^2 x}$

► $\frac{-\sin x}{1 + \sin^2 x}$

☐ $\frac{1}{1-\sin^2 x}$
☒ $\frac{\sin x}{1-\sin^2 x}$

Question No: 9 (Marks: 1) - Please choose one

$30^\circ =$ _____

☐ $\frac{\pi}{3}$
☐ $\frac{\pi}{4}$
☒ $\frac{\pi}{6}$

☐ $\frac{\pi}{2}$

Question No: 10 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}((c) \{h(x)\}) =$ _____

☒ 0

- ▶ $\frac{d}{dx}(h(x))$
- ▶ $\frac{d}{dx}(h(cx))$
- ☒ $c \frac{d}{dx}(h(x))$

Question No: 11 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\operatorname{cosec} x] = \underline{\hspace{2cm}}$$

- ▶ $\frac{1}{1 + \cos^2 x}$
- ▶ $\frac{-\cos x}{1 - \cos^2 x}$
- ☒ $\frac{-\cos x}{1 - \cos^2 x}$
- ▶ $\frac{1}{1 - \cos^2 x}$

Question No: 12 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- ▶ **Product**
- ▶ Sum
- ▶ Difference

► **Composition**

Question No: 13 (Marks: 1) - Please choose one

The power rule, $\frac{d}{dx}[x^n] = nx^{n-1}$ holds if n is _____

- An integer
- A rational number
- An irrational number

► **All of the above**

Question No: 14 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- **f is an increasing function.**
- f is a decreasing function.
- f is a constant function.

Question No: 15 (Marks: 1) - Please choose one

If $f''(x) > 0$ on an open interval (a, b) , then which of the following statement is correct?

- **f is concave up on (a, b) .**
- f is concave down on (a, b) .
- f is linear on (a, b) .

Question No: 16 (Marks: 1) - Please choose one

If

$f''(x) < 0$ on an open interval (a, b) then which of the following statement is correct?

- ▶ f is concave up on (a, b) .
- ▶ f is concave down on (a, b)
- ▶ f is linear on (a, b) .

Question No: 17 (Marks: 1) - Please choose one

If $x > 0$ then $\frac{d}{dx}[\ln x] =$ _____

- ▶ 1
- ▶ x
- ▶ $\frac{1}{x}$
- ▶ $\ln \frac{1}{x}$

Question No: 18 (Marks: 1) - Please choose one

If $b > 0$ then $\frac{d}{dx}[b^x] =$ _____

- ▶ 0
- ▶ xb^{x-1}
- ▶ $\ln b$
- ▶ $b^x \ln b$

Question No: 19 (Marks: 1) - Please choose one

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

▶ $\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$

▶ $\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$

▶ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$

▶ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$

Question No: 20 (Marks: 1) - Please choose one

What is the base of natural logarithm?

▶ 2.71

▶ 10

▶ 5

▶ Any real number

Question No: 21 (Marks: 1) - Please choose one

Let x_0 be critical points of the function f . Those critical points for which $f'(x_0) = 0$ are called _____ of f

► Local points

► End points

► **Stationary points**

Question No: 22 (Marks: 1) - Please choose one

$$\log_b a^r = \underline{\hspace{2cm}}$$

► $a \log_b r$

► **$r \log_b a$**

► $\frac{\log_b a}{\log_b r}$

►

► $\log_b a + \log_b r$

Question No: 23 (Marks: 1) - Please choose one

$$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$$

► $\log_b c$

- ▶ $1 - \log_b c$
- ☒ $-\log_b c$
- ▶ $1 + \log_b c$

Question No: 24 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

- ▶ $\log_b t$
- ▶ $1 - \log_b t$
- ▶ $1 + \log_b t$
- ☒ $-\log_b t$

Question No: 25 (Marks: 3)

If $f(x) = x^4 - 8x^2$, determine all relative extrema for the function. Using First Derivative Test.

Solution:

$$f = x^4 - 8x^2$$

$$f' = 4x^3 - 16x$$

$$f' = 0$$

$$4x^3 - 16x = 0$$

$$x(4x^2 - 16) = 0$$

$$x = 0$$

$$4x^2 - 16 = 0$$

$$x^2 = \frac{16}{4}$$

$$x^2 = 4$$

$$x = \pm 2$$

Relative extrema (0, ±2)....Ans

Question No: 26 (Marks: 5)

Differentiate $y = x^{-2}(4 + 3x^{-3})$

Solution:

$$y = 4x^{-2} + 3x^{-3} \cdot x^{-2}$$

$$= 4x^{-2} + 3x^{-5}$$

$$\frac{dy}{dx} = 4 \frac{d}{dx}(x^{-2}) + 3 \frac{d}{dx}(x^{-5})$$

$$= 4(-2)x^{-2-1} + 3 \frac{d}{dx} - 5x^{-5-1}$$

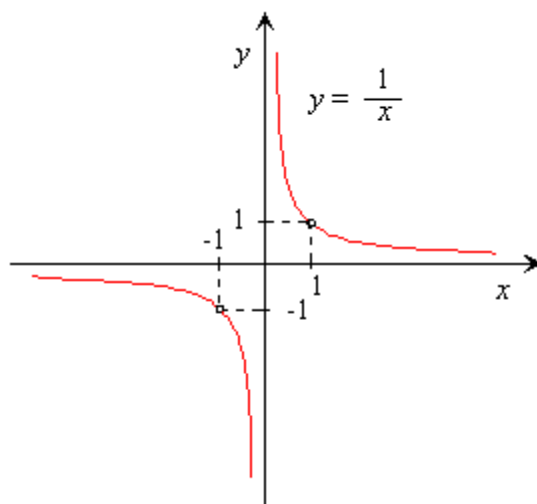
$$= -8x^{-3} + (-15x^{-6})$$

$$\frac{dy}{dx} = -8x^{-3} - 15x^{-6} \dots \dots \text{Ans}$$

Question No: 27 (Marks: 10)

Determine the intervals in which the graph of the function

$f(x) = \frac{1}{x}$
is concave upward or downward.



Solution:

$$f(x) = \frac{1}{x}$$
$$f'(x) = -\frac{1}{x^2}$$

X	1	2	3	4
---	---	---	---	---

F(x)	-1	-0.25	-0.11	-0.625

Conclusion:

$f'(x)$ is increasing when x is from $(0, \infty)$

so,

It is concave up.

$f'(x)$ is decreasing when x is from $(-\infty, 0)$

so,

It is concave down

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Question No: 1 (Marks: 1) - Please choose one

The average velocity of a body is V_{ave}

☒ $\frac{d_1 - d_0}{t_1 - t_0}$

☐ $\frac{t_1 - t_0}{f(t_0) - f(t_1)}$



☐ $\lim_{t_1 \rightarrow t_0} \frac{f(t_0) - f(t_1)}{t_1 - t_0}$



☐ None of these

Question No: 2 (Marks: 1) - Please choose one

Consider two functions $f(x) = x^3$ and $g(x) = (x+9)$ then $f \circ g(x) =$

☒ $(x+9)^3$

☐ $x+3$

☐ $x+9$

☐ None of these

Question No: 3 (Marks: 1) - Please choose one

Consider two function $f(x) = x^2$ and $g(x) = \sqrt{x}$ then $f \circ g(x) = \dots\dots\dots$

- ▶ x
- ▶ x^2
- ▶ $\sqrt{x}$
- ▶ None of these

Question No: 4 (Marks: 1) - Please choose one

Consider two functions $f(x) = 3\sqrt{x}$ and $g(x) = \sqrt{x}$ what is true about these functions

- ▶ $f(x).g(x) = 3x$
- ▶ $\frac{f(x)}{g(x)} = 3x$
- ▶ $f(g(x)) = 3x$
- ▶ None of these -correct

Question No: 5 (Marks: 1) - Please choose one

The centre and the radius of the circle $(x+5)^2 + (y-3)^2 = 16$ is

- ▶ **$(-5, 3), 4$**
- ▶ $(5, -3), 16$
- ▶ $(5, -3), 4$

- ▶ None of these

Question No: 6 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about

▶ **X-axis**

▶ Y-axis

▶ Origin

▶ None of these

Question No: 7 (Marks: 1) - Please choose one

The chain rule is used for two function f and g, if we have ----- of these functions

▶ Product

▶ Sum

▶ **Composition**

▶ None of these

Question No: 8 (Marks: 1) - Please choose one

function f is differentiable function if it is differentiable on the interval

▶ **$(-\infty, \infty)$**

A

- ▶ (a, ∞) where a is any negative integer
- ▶ $(0, \infty)$
- ▶ None of these

Question No: 9 (Marks: 1) - Please choose one

A function is said to be continuous function if the function is continuous on the interval

- ▶ $(-\infty, +\infty)$
- ▶ $(0, +\infty)$
- ▶ $(-\infty, 0)$
- ▶ None of these

Question No: 10 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

- ▶ 1
- ▶ 2
- ▶ 0
- ▶ 1/2

Question No: 11 (Marks: 1) - Please choose one

For any polynomial $P(x) = c_0 + c_1x + \dots + c_nx^n$ and any real number a
 $\lim_{x \rightarrow a} P(x) = c_0 + c_1a + \dots + c_na^n =$

▶ $P(a)$

▶ $P(a+1)$

▶ $P(a-1)$

▶ $\frac{1}{P(a)}$

▶

Question No: 12 (Marks: 1) - Please choose one

The no of x and y intercepts for the equation $y=1/x$

▶ Two x intercepts

▶ Two y intercepts

▶ No x and no y intercepts

▶ None of these

Question No: 13 (Marks: 1) - Please choose one

A

line is called a tangent line to the circle if it meets the circle at precisely

▶ One point

▶ Two points

▶ Infinite points

▶ None of these

Question No: 14 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

- ▶ Minima
- ▶ **Maxima**
- ▶ None of these

Question No: 15 (Marks: 1) - Please choose one

If the $\lim_{x \rightarrow a} f(x) = L$ then the inequality $(L - \varepsilon) < f(x) < L + \varepsilon$ holds in any subset of the interval

- ▶ **$(a - \delta, a) \cup (a, a + \delta)$**
- ▶ $(a - 1, a) \cup (a, a + 1)$
- ▶ $(a - \varepsilon, a) \cup (a, a + \varepsilon)$
- ▶ None of these

Question No: 16 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow 5} \frac{\sqrt{x+4}}{x^2+2} =$$

▶ 0

▶ ∞

▶ 1/9

▶ Limit doesn't exist

Question No: 17 (Marks: 2)

Show that $f(x) = x^2 - 3x + 1$ is a continuous function.

$$\lim_{x \rightarrow +\infty} (x^2 - 3x + 1) = +\infty$$

And

$$\lim_{x \rightarrow -\infty} (x^2 - 3x + 1) = +\infty$$

Question No: 18 (Marks: 2)

Find the range of function f defined by $f(x) = x^2 + 5$

Question No: 19 (Marks: 3)

Differentiate: $y = (\cos x)^{6x}$

Question No: 20 (Marks: 5)

Differentiate w.r.t. x by chain rule $y = \cos^2(x^3)$

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MIDTERM EXAMINATION

Fall 2009

MTH101- Calculus And Analytical Geometry

Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

▶ Minima

▶ Maxima

► None of these

Question No: 2 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

► Minima

► **Maxima**

► None of these

Question No: 3 (Marks: 1) - Please choose one

A line $x = x_0$ is called ----- for the graph of a function f if $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

► Horizontal asymptotes

► None of these

► **Vertical asymptotes**

Question No: 4 (Marks: 1) - Please choose one

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) = \underline{\hspace{2cm}}$

► $3x^7 + 2$

☒ $24x^7 + 2$

☐ $3x^9 + 2x^2$

☐ $24x^9 + 2x^2$

Question No: 5 (Marks: 1) - Please choose one

If

$y = \frac{1}{1-x}$ then $\frac{dy}{dx} =$

☐ 1

☐ -1

☒ $\frac{1}{(1-x)^2}$

☐ $\frac{-1}{(1-x)^2}$

Question No: 6 (Marks: 1) - Please choose one

If

$2x - y = -3$ then $\frac{dy}{dx} =$

☒ 2

☐ -2

☐ 0

☐ -3

Question No: 7 (Marks: 1) - Please choose one

If

$x^2 + y^2 = 9$ then $\frac{dy}{dx} =$

$\frac{x}{y}$



$\frac{-x}{y}$



$\frac{-y}{x}$



$\frac{y}{x}$



Question No: 8 (Marks: 1) - Please choose one

$\frac{d}{dx}[\sec x] =$ _____

$\frac{1}{1+\sin^2 x}$



$\frac{-\sin x}{1+\sin^2 x}$



$\frac{1}{1-\sin^2 x}$





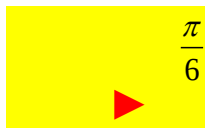
$$\frac{\sin x}{1 - \sin^2 x}$$

Question No: 9 (Marks: 1) - Please choose one

$$30^\circ = \underline{\hspace{2cm}}$$

 $\frac{\pi}{3}$

 $\frac{\pi}{4}$



$$\frac{\pi}{6}$$

 $\frac{\pi}{2}$

Question No: 10 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx}[f][g] =$$

- ▶ $\frac{[f'] [g] - [f] [g']}{g^2}$
- ▶ $[f'] [g']$
- ▶ $[f'] [g] + [f] [g']$
- ▶ $[f'] [g] - [f] [g']$

Question No: 11 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx} \left[\frac{f}{g} \right] =$$

- ▶ $\frac{[g] [f'] - [f] [g']}{g^2}$
- ▶ $\frac{[g'] [f] - [f'] [g]}{g^2}$
- ▶ $\frac{[g] [f'] - [f] [g']}{f^2}$
- ▶ $\frac{[g'] [f] - [f'] [g]}{f^2}$

Question No: 12 (Marks: 1) - Please choose one

If a function g is differentiable at a point x and a function f is differentiable at a point $g(x)$, then the _____ is differentiable at point x .

► **Composition** (f o g)

- Quotient (f / g)
- Product (f . g)
- Sum (f + g)

Question No: 13 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- Product
- Sum
- Difference

► **Composition**

Question No: 14 (Marks: 1) - Please choose one

The power rule, $\frac{d}{dx}[x^n] = nx^{n-1}$ holds if n is _____

- An integer
- A rational number
- An irrational number

► **All of the above**

Question No: 15 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2

denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

▶ f is an increasing function.

- ▶ f is a decreasing function.
- ▶ f is a constant function.

Question No: 16 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- ▶ f is an increasing function.

▶ f is a decreasing function.

- ▶ f is a constant function.

Question No: 17 (Marks: 1) - Please choose one

If $f''(x) > 0$ on an open interval (a, b) , then which of the following statement is correct?

▶ f is concave up on (a, b)

- ▶ f is concave down on (a, b) .
- ▶ f is linear on (a, b) .

Question No: 18 (Marks: 1) - Please choose one

If $x > 0$ then $\frac{d}{dx}[\ln x] =$ _____

▶ 1

▶ x

▶ $\frac{1}{x}$

▶ $\ln \frac{1}{x}$

Question No: 19 (Marks: 1) - Please choose one

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

$$\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$$



$$\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$$



$$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$$



$$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$$



Question No: 20 (Marks: 1) - Please choose one

What is the base of natural logarithm?

▶ 2.71

► 10

► 5

► Any real number

Question No: 21 (Marks: 1) - Please choose one

If

we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} =$ _____

► $\frac{-x}{y}$

► $\frac{x}{y}$

► $\frac{-y}{x}$

► None of these

Question No: 22 (Marks: 1) - Please choose one

_____ $\log_b a^r =$ _____

► $a \log_b r$

► $r \log_b a$

$$\frac{\log_b a}{\log_b r}$$



▶ $\log_b a + \log_b r$

Question No: 23 (Marks: 1) - Please choose one

$$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$$

▶ $\log_b c$

▶ $1 - \log_b c$

▶ $-\log_b c$

▶ $1 + \log_b c$

Question No: 24 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

▶ $\log_b t$

▶ $1 - \log_b t$

▶ $1 + \log_b t$

▶ $-\log_b t$

Question No: 25 (Marks: 3)

Differentiate: $y = x^{\sqrt{x}} e^{5x+6}$

Question No: 26 (Marks: 5)

Differentiate $y = (x^3 + 7x - 1)(5x + 2)$

Question No: 27 (Marks: 10)

The derivative of a continuous function is given .Find all critical points and determine whether a relative maximum, relative minimum or neither occur there

$$f'(x) = 2\sin^3 x - \sin^2 x \quad ; \quad 0 < x < 2\pi$$

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MIDTERM EXAMINATION

Spring 2010

MTH101- Calculus And Analytical Geometry

Time: 60 min

Marks: 40

Question No: 1 (Marks: 1) - Please choose one

$30^0 =$ _____

- ▶ $\frac{\pi}{3}$
- ▶ $\frac{\pi}{4}$
- ▶ $\frac{\pi}{6}$
- ▶ $\frac{\pi}{2}$

Question No: 2 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then which of the following statement is correct?

- ▶ f is a decreasing function
- ▶ f is an increasing function
- ▶ f is a constant function

Question No: 3 (Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

 $\pm \frac{k\pi}{2} (k = 1, 3, 5, \dots)$

 $\pm \frac{k\pi}{2} (k = 2, 4, 6, \dots)$

 $\pm \frac{k\pi}{2} (k = 1, 2, 3, 4, 5, 6, \dots)$



Question No: 4 (Marks: 1) - Please choose one

$\lim_{x \rightarrow \infty} (-2x) =$

 -2

 0

 2

 **Does not exist**

Question No: 5 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$\frac{d}{dx}[f][g] =$

- ▶ $\frac{[f'] [g] - [f] [g']}{g^2}$
- ▶ $[f'] [g']$
- ▶ $[f'] [g] + [f] [g']$
- ▶ $[f'] [g] - [f] [g']$

Question No: 6 (Marks: 1) - Please choose one

The solution set of the inequality $|x+4| \geq 2$ is

- ▶ $(-\infty, -6] \cup [2, +\infty)$
- ▶ **None of these**
- ▶ $(-\infty, 6] \cup [-2, +\infty)$
- ▶ $(-\infty, -6] \cup [-2, +\infty)$

Question No: 7 (Marks: 1) - Please choose one

A line is called a tangent line to the circle if it meets the circle at precisely

- ▶ **One point**
- ▶ Two points
- ▶ Infinite points

Question No: 8 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$

then which of the following statement is correct?

▶ f is an increasing function.

▶ f is a decreasing function.

▶ f is a constant function.

Question No: 9 (Marks: 1) - Please choose one

If

$f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

▶ $3x^7 + 2$

▶ $24x^7 + 2$

▶ $3x^9 + 2x^2$

▶ $24x^9 + 2x^2$

Question No: 10 (Marks: 1) - Please choose one

π is anumber

▶ Integer

▶ Rational

▶ Irrational

▶ Natural

Question No: 11 (Marks: 1) - Please choose one

The set $\{x : a \leq x \leq b\}$ can be written in the form of interval

► (a,b)

► (a,b]

► [a,b]

Question No: 12 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx} \left[\frac{f}{g} \right] =$$

► $\frac{[g][f'] - [f][g']}{g^2}$

► $\frac{[g'][f] - [f'][g]}{g^2}$

► $\frac{[g][f'] - [f][g']}{f^2}$

► $\frac{[g'][f] - [f'][g]}{f^2}$

Question No: 13 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about -----axis

► X-axis

► Y-axis

► Origin

Question No: 14 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow -7} \frac{x^2 - 49}{x + 7}$$

▶ -14

▶ 0

▶ ∞

▶ Limit does not exist

Question No: 15 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

▶ Product

▶ Sum

▶ Difference

▶ **Composition**

Question No: 16 (Marks: 1) - Please choose one

The $\lim_{x \rightarrow a} f(x) = \dots\dots\dots$ where $f(x) = k$ (k is a constant)

▶ k+2

▶ k+1

▶ **k**

Question No: 17 (Marks: 1) - Please choose one

For any number $\varepsilon > 0$ if we can find an open interval (x_0, x_1) on the x-axis containing the point "a" such that $L - \varepsilon < f(x) < L + \varepsilon$ for each x in (x_0, x_1) except the possible $x = a$ then we say $\lim_{x \rightarrow a} f(x) =$ -----

- ☒ L
- ☐ $L - \varepsilon$
- ☐ $L + \varepsilon$
- ☐ $L + 1$

Question No: 18 (Marks: 1) - Please choose one

If $2x - y = -3$ then

$$\frac{dy}{dx} =$$

- ☒ 2
- ☐ -2
- ☐ 0
- ☐ -3

Question No: 19 (Marks: 1) - Please choose one

The graph of the equation $y = x^2 - 4x + 5$ will represent

☒ Parabola

- ▶ Straight line
- ▶ two straight lines

- ▶ Ellipse

Question No: 20 (Marks: 1) - Please choose one

The equation of line of the form $y - y_1 = m(x - x_1)$ is known as

▶ **Point-slope form**

- ▶ Two points form
- ▶ Intercepts form
- ▶ Slope intercept form

Question No: 21 (Marks: 2)

If $y = 5 \cos(x^2 + 1)$ Find dy/dx by using "The chain rule".

Question No: 22 (Marks: 2)

Prove that $\lim_{x \rightarrow 1} (x + 5) = 6$, using the definition of limit.

Question No: 23 (Marks: 3)

Find an equation of the tangent line to the curve

$$y = \frac{2}{x^2 + x}$$

at the point where $x = 1$

Question No: 24 (Marks: 3)

Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{3x}$

Question No: 25 (Marks: 5)

Find all critical points of $f(x) = x^4 - 4x^3 + 4x^2$

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Question No: 1 (Marks: 1) - Please choose one

The set $\{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is known as set of

- ▶ Natural numbers
- ▶ **Integers**
- ▶ Whole numbers
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

$$h(x) = \frac{1}{(x-2)(x-4)}$$

The domain of the function is

- ▶ **$(-\infty, 2) \cup (2, 4) \cup (4, +\infty)$**

- ▶ $(-\infty, 2] \cup \{2, 4\} \cup \{4, \infty)$
- ▶ $(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$ _
- ▶ All of these are incorrect

Question No: 3 (Marks: 1) - Please choose one

If the $\lim_{x \rightarrow a} f(x) = L$ then the inequality $(L - \varepsilon) < f(x) < L + \varepsilon$ holds in any subset of the interval

- ▶ $(a - \delta, a) \cup (a, a + \delta)$
- ▶ $(a - 1, a) \cup (a, a + 1)$
- ▶ $(a - \varepsilon, a) \cup (a, a + \varepsilon)$
- ▶ **None of these**

Question No: 4 (Marks: 1) - Please choose one

$L - \varepsilon < f(x) < L + \varepsilon$ Can be written as

- ▶ $|f(x) - L| < \varepsilon$
- ▶ $|f(x) - L| > \varepsilon$
- ▶ $|f(x) - L| < \varepsilon + 1$
- ▶ None of these

Question No: 5 (Marks: 1) - Please choose one

If a function satisfies the conditions

$f(c)$ is defined

$$\lim_{x \rightarrow c^+} f(x)$$

Exists

$$\lim_{x \rightarrow c^+} f(x) = f(c)$$

Then the function is said to be

► **Continuous at c**

- Continuous from left at c
- Continuous from right at c
- None of these

Question No: 6 (Marks: 1) - Please choose one

$$\frac{d}{dx} [\sec x] = \text{-----}$$

► $\frac{\sin x}{1 - \sin^2 x}$

► $\frac{-\sin x}{1 - \sin^2 x}$

► $\frac{1}{1 - \sin^2 x}$

►

► None of these

Question No: 7 (Marks: 1) - Please choose one

$$\log_b ac = \text{-----}$$

► $\log_b a + \log_b c$

► $\log_a b + \log_c b$

► $\log_{a+c} b$

► None of these

Question No: 8 (Marks: 1) - Please choose one

$$\log_b a^r = \text{-----}$$

► $a \log_b r$

► $r \log_b a$

► $b \log_a r$

► None of these

Question No: 9 (Marks: 1) - Please choose one

$f''(x) < 0$ on an open interval (a,b) then f is ----- on (a,b)

If

► None of these

- ▶ Concave up
- ▶ **Concave down**
- ▶ Closed

Question No: 10 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

- ▶ **Minima**
- ▶ Maxima
- ▶ None of these

Question No: 11 (Marks: 1) - Please choose one

line is called a tangent line to the circle if it meets the circle at precisely

A

- ▶ **One point**
- ▶ Two points
- ▶ Infinite points
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at and radius

- ▶ **$(-4,1), \sqrt{6}$**
- ▶ $(-4,1), 6$

- ▶ $(-4, -1), \sqrt{6}$
- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

The $\lim_{x \rightarrow a} f(x)$ where $f(x) = k$ (k is a constant) is equal to

- ▶ $k+2$
- ▶ $k+1$
- ▶ **k**
- ▶ kf

Question No: 14 (Marks: 1) - Please choose one

For any polynomial $P(x) = c_0 + c_1x + \dots + c_nx^n$ and any real number a

$\lim_{x \rightarrow a} P(x) = c_0 + c_1a + \dots + c_na^n =$

- ▶ $P(a)$
- ▶ $P(a+1)$
- ▶ $P(a-1)$
- ▶ $\frac{1}{P(a)}$
- ▶

Question No: 15 (Marks: 1) - Please choose one

Polynomials are always Function

- ▶ **Continuous**
- ▶ Discontinuous

Question No: 16 (Marks: 1) - Please choose one

$$\frac{D}{Dx}[dh(x)] = \text{-----}$$

where d is a constant

- ▶ $dh(x)$
- ▶ $dh'(x)$
- ▶ 0
- ▶ None of these

Question No: 17 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about

- ▶ X-axis
- ▶ Y-axis
- ▶ Origin
- ▶ None of these

Question No: 18 (Marks: 1) - Please choose one

Consider two function $f(x) = 3\sqrt{x}$ and $g(x) = \sqrt{x}$ what is true about these functions

- ▶ $f(x).g(x) = 3x$
- ▶ $\frac{f(x)}{g(x)} = 3x$
- ▶ $f(g(x)) = 3x$
- ▶ None of these

Question No: 19 (Marks: 1) - Please choose one

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The formula is called With respect to x of the function f

► **Derivative**

- Slope
- Tangent
- None of these

Question No: 20 (Marks: 1) - Please choose one

$$\frac{d}{dx} \left(\frac{f}{g} \right)$$

Suppose that f and g are differentiable function of x then

► $\frac{g \cdot f' - f \cdot g'}{g^2}$

► $\frac{g \cdot f' + f \cdot g'}{g^2}$

► $\frac{g \cdot f' - f \cdot g'}{g}$

- None of these

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Question No: 1 (Marks: 1) - Please choose one

Consider two function $f(x) = x^2$ and $g(x) = \sqrt{x}$ then $f \circ g(x) =$

- ☒ x
- ☐ x^2
- ☐ $\sqrt{x}$
- ☐ None of these

Question No: 2 (Marks: 1) - Please choose one

Natural domain of $\frac{(x^2 - 4)}{(x - 2)}$ is

- ☒ $(-\infty, 2) \cup (2, +\infty)$
- ▶ $(-\infty, 2)$
- ▶ $(-\infty, 0)$
- ▶ None of these

Question No: 3 (Marks: 1) - Please choose one

The solution of the inequality $|x - 3| < 3$ is

- ▶ $(-1, 7)$
- ▶ $(1, 7)$
- ▶ $(1, -7)$
- ▶ **None of these**

Question No: 4 (Marks: 1) - Please choose one

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is of x

- ▶ relation
- ▶ none of these
- ▶ **function**
- ▶ not function

Question No: 5 (Marks: 1) - Please choose one

The set of all points in the coordinate plane which are at a fixed distance away from a given fixed point represents

- ▶ Parabola
- ▶ Straight line
- ▶ **Circle**
- ▶ None of these

Question No: 6 (Marks: 1) - Please choose one

Let L_1 and L_2 be non vertical lines with slopes m_1 and m_2 , respectively both the lines are perpendicular if and only if

- ☒ $m_1(-m_2) = 1$
- ▶ $m_1 m_2 = -1$
- $m_1 = -\frac{1}{m_2}$
- ▶
- ▶ **All of these**

Question No: 7 (Marks: 1) - Please choose one

The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at and radius

- ☒ $(-4,1), \sqrt{6}$
- ▶ $(-4,1), 6$
- ▶ $(-4,-1), \sqrt{6}$
- ▶ None of these

Question No: 8 (Marks: 1) - Please choose one

The base of the natural logarithm is

▶ 2.71

▶ 10

▶ 5

▶ None of these

Question No: 9 (Marks: 1) - Please choose one

$$\frac{d}{dx}[f(g(x))]=$$

▶ $f'(g(x)).g'(x)$

▶ $f'(g(x))+g'(x)$

▶ $f'(g(x)).f'(x)$

▶ None of these

Question No: 10 (Marks: 1) - Please choose one

If $\frac{dy}{dx} =$

$y=f(g(h(x)))$ and $u=g(h(x))$ and $v=h(x)$ then

▶ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$

$$\frac{dy}{du} + \frac{du}{dv} + \frac{dv}{dx}$$



$$\frac{dy}{dx} \cdot \frac{du}{dv} \cdot \frac{dv}{du}$$



▶ None of these

Question No: 11 (Marks: 1) - Please choose one

The $\tan(x)$ is discontinuous at the points where

▶ $\cos(x) = 0$

▶ $\sin(x) = 0$

▶ $\tan(x) = 0$

▶ None of these

Question No: 12 (Marks: 1) - Please choose one

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Equals to

▶ 1

▶ 2

▶ 3

▶ 0

Question No: 13 (Marks: 1) - Please choose one

Both $\sin(x)$ and $\cos(x)$ have the same limit and function value at $x=0$ so both are at $x=0$

- ▶ Continuous
- ▶ **Discontinuous**
- ▶ Linear
- ▶ None of these

Question No: 14 (Marks: 1) - Please choose one

For a function $f(x)$ to be continuous on interval (a,b) the function must be continuous

- ▶ **At all point in (a,b)**
- ▶ Only at a and b
- ▶ At mid point of a and b
- ▶ None of these

Question No: 15 (Marks: 1) - Please choose one

is called

π

- ▶ An integer
 - ▶ **A rational number**
 - ▶ An irrational number
 - ▶ A natural number
-

Question No: 16 (Marks: 1) - Please choose one

The graph of the equation $y = x^2 - 4x + 5$ will represent

▶ **Parabola**

- ▶ Ellipse
- ▶ Straight line
- ▶ Two straight line

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1) What is Calculus??

the study of the continuous rates of the change of quantities. It is the study of how various quantities change with respect to other quantities. For example, one would like to know how distance changes with respect to time, or how time changes w.r.t speed, or how water flow changes w.r.t time etc

2) the simplest numbers are the ____?

- a) Real numbers
- b) Odd numbers
- c) Complex numbers
- d) Natural numbers

3) Why natural numbers are called natural numbers?

Because they were first which have cross paths with human intellect. we count things with them. Our ancestors used these numbers first to count, and they came to us naturally and called as natural numbers.

4) The natural numbers form a subset of a larger class of numbers called the ____.

- a) Subset of natural no.
- b) Integers
- c) Set
- d) Both a and b

5) The collection of things with reference of numbers is called

- a) Information
- b) Counting
- c) Set
- d) Integer

6) If A is the subset of B then

- a) $A \subseteq B$
- b) $B \subseteq A$
- c) $B \subseteq B$
- d) $A \subseteq A$

7) We have two sets $A=\{1, 2, 3, 4, 5\}$ & $B=\{2, 3\}$ what is correct option belongs to these two sets.

- a) A is subset of B
- b) B is subset of A
- c) $B \subseteq A$
- d) Both b and c

8) ..., -4, -3, -2, -1, 0, 1, 2, 3, 4,... these are ____?

- a) Whole numbers
- b) Negative numbers
- c) Natural numbers
- d) None of these

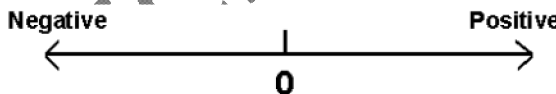
9) $x+2 = 0$. The solution is ____?

- a) $x = -2$
- b) $x = +2$
- c) $x = 1$
- d) $x = -1$

10) The integers in turn are a subset of a still larger class of numbers called the

- a) rational number

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- b) natural number
c) irrational numbers
d) real numbers
- 11) If x is different from zero, this equation is
a) Contradictory
b) Satisfied
c) Unsatisfied
d) Defined
- 12) _____ was an ancient Greek philosopher and mathematician who studied the properties of numbers.
a) Robert Hook
b) Pythagoras
c) H. Methelda
d) Both a and b
- 13) Pythagoras concluded that the size of a physical quantity must consist of a _____ of an additional unit.
a) whole number + some fraction m/n
b) Negative number + some fraction m/n
c) natural number + some fraction m/n
d) real number + some fraction m/n
- 14) Using geometric methods, Pythagoras showed that the hypotenuse of the right triangle with base and opposite side equal to _____ cannot be expressed as the ratio of integers
a) 0
b) 1
c) -1
d) Both a and b
- 15) The rational and irrational numbers together comprise a larger class of numbers, called
a) REAL NUMBERS
b) WHOLE NUMBERS
c) NATURAL NUMBERS
d) Even numbers
- 16)  in this diagram origin is:
a) Negative terminal
b) Positive terminal
c) 0
d) Both a and b
- 17) $24,0,53\pi \geq \leq -3$ is a _____ inequality
a) Correct
b) Incorrect
c) Both in different origin
- 18) $A = \{1,2,3,4\}$, $B = \{1,2,3,4,5,6,7\}$, then, $A \cup B =$
a) $\{1,1,2,3,4,5,6,7\}$
b) $\{1,2,3,4,4,5,6,7\}$
c) $\{1,2,3,4,5,6,6,7\}$
d) $\{1,2,3,4,5,6,7\}$
- 19) $A = \{1,2,3,4\}$, $B = \{1,2,3,4,5,6,7\}$, then $A \cap B =$
a) $\{1,2,3,4,5,6\}$
b) $\{1,2,3,4\}$
c) $\{1,2,3,4,5,6,7\}$

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d) $\{1,2,3,4,5\}$

20) The absolute value or magnitude of a real number a is denoted by $|a|$ and is defined by

a) $|a| = \begin{cases} a & \text{if } a > 0 \\ -a & \text{if } a < 0 \end{cases}$ that is, a is non-negative

b) $|a| = \begin{cases} -a & \text{if } a < 0 \\ a & \text{if } a > 0 \end{cases}$ that is, a is positive

c) Both a and b

d) None of these

21) Solve $|x-3|=4$

Ans: If we depend upon $x-3$ is positive or negative, the equation $|x-3|=4$ can be written as

$x-3=4$ ($- + = -$) then it will be written as $x=7$ mean in other words ($7-3=4$)

$x-3=-4$ ($- - = +$) then it will be written as $x=-1$ mean in other words ($-1 & -3 = -4$)

22) What is Relationship between Square Roots and Absolute Values?

Ans: A number whose square is a is called a square root of a . The absolute value or magnitude of a real number a is denoted by $|a|$ and is defined by $|a| = \begin{cases} a & \text{if } a > 0 \\ -a & \text{if } a < 0 \end{cases}$ that is, a is non-negative & $\{-a$ if $a < 0$ that is, a is positive $\}$

23) $a, \sqrt{a^2} = |a|$ is used for any

a) Real number

b) Absolute value

c) Natural number

d) Whole number

24) What is distance formula?

If A and B are points on a coordinate line with coordinates a and b , respectively, then the distance d between A and B is $d = |b-a|$. This formula provides a useful geometric interpretation.

25) What do you know about triangle inequality?

If a and b are any real numbers, then $|a+b| \leq |a| + |b|$ it is called Triangle Inequality.

26) Which is the best example of Pythagoras' conclusion

a) $1/2 = 0.500000... = 0.5$

b) $1/3 = 0.33333...$

c) $1/4 = 0.44444... = 0.4$

d) All of these

27) What is a plane?

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A PLANE is just the intersection of two COORDINATE lines at 90 degrees. It is technically called the COORDINATE PLANE

- 28) To plot a point $P(a, b)$ means to locate the point with _____ in a coordinate plane
- a) coordinates (a, c)
 - b) coordinates (a, b)
 - c) coordinates (b, c)
 - d) coordinates (b, a)
- 29) The COORDINATE PLANE and the ordered pairs we just discussed is together known as the
- a) RECTANGULAR COORDINATE SYSTEM
 - b) RECTANGULAR GRAPH SYSTEM
 - c) RECTANGULAR PLAN SYSTEM
 - d) Both a and b
- 30) In a rectangular coordinate system the coordinate axes divide the plane into four regions called
- a) Planes
 - b) Exis
 - c) quadrants.
 - d) Both a and c
- 31) Upper left quadrant is called
- a) (I)
 - b) (II)
 - c) (III)
 - d) (IV)
- 32) Lower right quadrant is called
- a) (I)
 - b) (II)
 - c) (III)
 - d) (IV)
- 33) What signs are written with 1st quadrant.
- a) $(-, +)$
 - b) $(+, +)$
 - c) $(+, -)$
 - d) $(-, -)$

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34) $6x - 4y = 10$
solution is?

- a) $6 - 4 = 10$
- b) $6(3) - 4(2) = 10$
- c) $6(2) - 4(0) = 18$ is not 10
- d) $6(0) - 4(0) = 10$

35) The _____ of an equation in two variables x and y is the set of all points in the xy -plane whose coordinates are members of the solution set of the equation.

- a) GRAPH
- b) PLANE
- c) NUMBERS
- d) EXIS

36) When a graph is obtained by plotting points, whether by _____ there is no guarantee that the resulting curve has the correct shape

- a) Hand
- b) Calculator
- c) Computer
- d) All of these

37) The number a is called an x -intercept of the graph and the number b a y -intercept then:

- a) x -axis $= (a, 0)$, y -axis $= (0, b)$
- b) x -axis $= (a, 1)$, y -axis $= (1, b)$
- c) x -axis $= (a, 1)$, y -axis $= (0, b)$
- d) x -axis $= (a, 0)$, y -axis $= (1, b)$

38) (a) $3x + 2y = 6$

(b) $x = y^2 - 2y$

(c) $y = 1/x$

$$3x + 2y = 6$$

***x*-intercepts**

Set $y = 0$ and solve for x

$$3x = 6 \quad \text{or} \quad x = 2$$

$$3x + 2y = 6$$

***y*-intercepts**

Set $x = 0$ and solve for y

$$2y = 6 \quad \text{or} \quad y = 3$$

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Page 5

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$$y = 1/x$$

x-intercepts

Set $y = 0$

$$1/x = 0 \Rightarrow x \text{ is undefined}$$

No x-intercept

y-intercepts

Set $x = 0$

$$y = 1/0 \Rightarrow y \text{ is undefined}$$

No y-intercept

39) _____ is at the heart of many mathematical arguments concerning the structure of the universe, and certainly symmetry plays an important role in applied mathematics and engineering fields

- a) Quadrant
- b) Plane
- c) line
- d) Symmetry

40) Slope of a hill is the _____ of its rise to its run:

- a) Ratio
- b) Equation
- c) Rate
- d) Time

41) Rising slope m is sometimes called the **rate of change of y with respect to**

- a) Y along axis
- b) X along axis
- c) Y along axis
- d) X along the line

42) slope of a line can be

- a) positive
- b) negative
- c) zero
- d) All of these

43) For a line L not parallel to the x -axis, the angle of _____ is the smallest angle measured counterclockwise from the direction of the positive x -axis to L

- a) Incidence
- b) Reflection
- c) quadrination
- d) inclination

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Page 6

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44) The line with y-intercept b and slope m is given by the equation $y = mx + b$ called ____

- a) **slope-intercept form** of the line
- b) **plane-intercept form** of the slope
- c) **angle-intercept form** of the line
- d) **angle-intercept form** of the slope

45) Where A , B and C are constants and A and B are not both zero, is called a ____-degree equation in x and y

- a) **First**
- b) Second
- c) Third
- d) Fourth

46) We use the equations of lines called _____ to study motions

- a) Coordinate equations
- b) **Linear equations**
- c) **Negative equations**
- d) **positive equations**

47) The distance d between two points (x_1, y_1) and (x_2, y_2) in a coordinate plane is given by

- a) $d = \sqrt{|x_2 - x_1|^2 + |y_2 - y_1|^2}$
- b) $d = \sqrt{|y_2 - y_1|^2 - |x_2 - x_1|^2}$ and is a
- c) $d = \sqrt{|x_2 - x_1|^2 - |y_2 - y_1|^2}$
- d) $d = \sqrt{|y_2 - y_1|^2 + |x_2 - x_1|^2}$

48) The distance between two points P_1 and P_2 in a coordinate plane is commonly denoted by

- a) $d(P_1, P_2)$
- b) $d(P_2, P_1)$
- c) $d(P_2, P_2)$
- d) **both a and b**

49) The midpoint of the line segment joining two points (x_1, y_1) and (x_2, y_2) in a coordinate plane is

- a) **$(X, y) = (1/2(x_1 + x_2), 1/2(y_1 + y_2))$**
- b) $(y, y) = (1/2(x_1 + x_2), 1/2(y_1 + y_2))$
- c) $(X, x) = (1/2(x_1 + x_2), 1/2(y_1 + y_2))$
- d) $(X, y) = (1/2(y_1 + y_2), 1/2(y_1 + y_2))$

50) This is called the standard form of the equation of circle.

- a) $(x - x)^2$

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Page 7

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b) $(y-y)^2$

c) $(x-x)^2+(y-y)^2=r^2$

d) $(x-x)^2-(y-y)^2=r^2$

51) X coordinates of the centre is

a) X nod

b) Y nod

c) 0

d) Both a and b

52) Y coordinates of the centre is

a) X nod

b) Y nod

c) 0

d) Both a and b

53) Radius is denoted by

a) Half of the circle

b) Inner point of the circle

c) By square of r

d) $R=x+y$

54) The circle $x^2+y^2=1$, which is centered at the origin and has radius 1, is of special importance; it is called the :

a) unit circle

b) mid point

c) radius

d) center

55) what are degenerated cases of the circle?

Ans: $k>0$, $k=0$, $k<0$

56) describe a graph of $(x-1)^2 + (y+4)^2 = -9$

Ans: There are no real values of x and y that will make the left side of the equation negative. Thus, the solution set of the equation is empty, and the equation has no graph.

57) The x-coordinate of the vertex of the parabola can be found by the following formula

a) $Y=-b/2a$

b) $X = -b/2a$

c) $Y=b/2a$

d) $X=b/2a$

58) To find these intercepts we set $y=0$ to obtain

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a) $x^2 - 2x - 2 = 0$

b) $x^2 - 2x - 2 = 1$

c) $x^2 - 2x - 2 = 2$

d) $x^2 - 2x - 2 = 3$

59) the height of the graph of a quadratic is maximum or minimum, depending on whether the graph opens

a) UP

b) DOWN

c) LONG

d) Both a and b

60) The area A of a circle depends on its ____ by the formula $a = \pi r^2$

a) radius

b) centre

c) height

d) breath

61) velocity is a _____ of time

a) SPEED

b) LENGTH

c) AREA

d) FUNCTION

62) "Area is a FUNCTION of

a) Time

b) Speed

c) Radius

d) Velocity

63) In the _____, Swiss mathematician Euler introduced the notation which we mean $y = f(x)$.

a) 1665

b) 1800

c) 1777

d) 1700

64) What Operations can be applied on Functions?

- Like numbers, functions can be OPERATED upon
- Functions can be added
- Functions can be subtracted
- Functions can be multiplied

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- Functions can be divided
- Functions can be COMPOSED with each other

65) We have $f(x)=x^2$, $g(x)=x$ if we want to add these two functions then which of the following option is correct?

- a) $f(x)+x^2 = g(x)+x$
- b) $f(x)+x^2 = g(x)=x$
- c) $f(x)=x^2 = g(x)+x$
- d) $f(x)+g(x)=x^2+x$

66) What are products of addition, subtraction, multiplication and division of these two functions $f(x)=x$, $g(x)=x$

$$\begin{aligned}(f+g)(x) &= f(x)+g(x) \\ (f-g)(x) &= f(x)-g(x) \\ (f \cdot g)(x) &= f(x) \cdot g(x) \\ \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}\end{aligned}$$

these are $f+g$, $f-g$, $f \cdot g$ and f/g

67) solve this $f(x)=1+\sqrt{x-2}$ $g(x)=x-1$

Handwritten solution for problem 67:

$$\begin{aligned}f(x) &= 1 + \sqrt{x-2} & g(x) &= x-1 \\ \text{by addition} \\ (f+g)(x) &= f(x) + g(x) \\ &\quad \downarrow x \text{ is common} \\ &= f(x) + g(x) \\ &= 1 + \sqrt{x-2} + x - 1 \\ &= +1 \text{ \& } -1 \\ &= +\sqrt{x-2} + x \\ &= x + \sqrt{x-2}\end{aligned}$$

Domains

$$\begin{aligned}f &= (2, +\infty) \\ g &= (-\infty, +\infty) \\ f+g &= (2, +\infty) \cap (-\infty, +\infty) \\ &= (2, +\infty)\end{aligned}$$

68) What is the domain of $3x$?

- a) $-\infty, -\infty$
- b) $+\infty, +\infty$
- c) $-\infty, +\infty$
- d) $+\infty, -\infty$

69) _____ operation has no analog with the arithmetic operations?

- a) COMPOSITION
- b) ADDITION
- c) SUBTRACTION
- d) MULTIPLICATION

70) When two functions are composed, ONE is assigned as a _____ to the independent variable of the other

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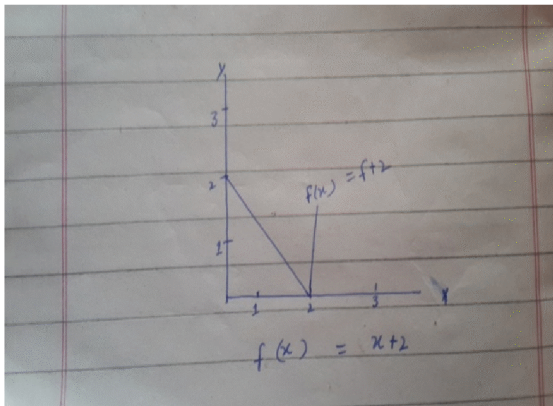
- a) Number
- b) Zero
- c) Value
- d) Both a and c

Function	g(x) Inside	f(x) Outside	composition
$(x^2+1)^{10}$	x^2+1	x^{10}	$(x^2+1)^{10}=f(g(x))$
$\sin^3x$	$\sin x$	x^3	$\sin^3x=f(g(x))$
$1/(x+1)$	$x+1$	$1/x$	$1/(x+1) = f(g(x))$
$\tan(x^5)$	x^5	$\tan x$	$\tan(x^5)=f(g(x))$

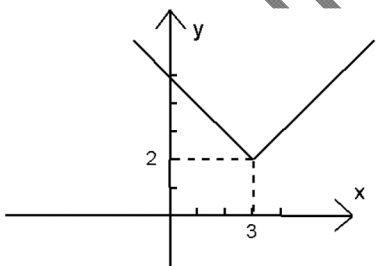
71) Define a graph?

A graph of an equation is just the points on the xy- plane that satisfy the equation

72) Sketch the graph of $F(x)=x+2$



73) Sketch the graph of $y=f(x)=|x-3|+3$



graph of $f(x) = |x - 3| + 2$

74) If $f(x)$ is MULTIPLIED by a POSITIVE constant c , then which of the following geometric effects take place?

- a) The graph of $f(x)$ is COMPRESSED vertically if $0 < c < 1$
- b) The graph of $f(x)$ is STRETCHED vertically if $c > 1$
- c) This is called VERTICAL Scaling by a factor of c
- d) All of these

75) A graph in the plane is the graph of a function if and only if _____ line intersects the graph more than once.

- a) VERTICAL
- b) NO VERTICAL
- c) PERPENDICULAR
- d) NON-PERPENDICULAR

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- 76) Traditionally, the Calculus that comes out of the tangent problem is called
- a) MULTIPLYCATIONAL CALCULUS
 - b) NON-DIFFERENTIAL CALCULUS
 - c) DIFFERENTIAL CALCULUS
 - d) ALL OF THESE
- 77) Calculus that comes out of the area problem is called
- a) DIFFERENTIAL CALCULUS
 - b) NON-DIFFERENTIAL CALCULUS
 - c) INTEGRAL CALCULUS
 - d) MULTIPLYCATIONAL CALCULUS
- 78) In geometry, a line is called ____ to a circle if it meets the circle at exactly one point
- a) coordinates
 - b) negative line
 - c) positive line
 - d) tangent
- 79) Consider a point P on a curve in the xy-plane. Let Q be another point other than P on the curve. Draw a line through P and Q to get what is called the ____ line for the curve. Now move the Point Q toward P. The Secant line will rotate to a "limiting" position as Q gets closer and closer to P. The line that will occupy this limiting position will be called the ____ line at P
- a) TANGENT, SECANT
 - b) SECANT, TANGENT
 - c) TANGENT, TANGENT
 - d) SECANT, SECANT
- 80) If we let our rectangles increase in number, then the approx will be better and the result will be getting as ____ on the number of rects.
- a) LIMITING value
 - b) Non- LIMITING value
 - c) Tangent
 - d) coordinates
- 81) ____ are basically a way to study the behavior of the y-values of a function in response to the x-values as they approach some number or go to infinity
- a) Limits
 - b) Tangents
 - c) Circles
 - d) lines
- 82) Consider $f(x) = \frac{\sin(x)}{x}$ where x is in
- a) Limits
 - b) Brackets
 - c) Radians
 - d) All of these
- 83) Remember that PI radians = ?
- a) 150 degree
 - b) 180 degrees
 - c) 0 degree
 - d) 90 degree

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84) $f(x)$ is not defined at $x =$

- a) 0
- b) 90
- c) 80
- d) 180

85) What happens if you get very close to $x = 0$?

Ans : We can get close to 0 from the left of 0, and from the right of 0. x can approach 0 along the negative x -axis means from the left. x can approach 0 along the positive x -axis means from the right. From both sides we get REALLY close to 0, but not equal to it. This getting really close is called the LIMITING process.

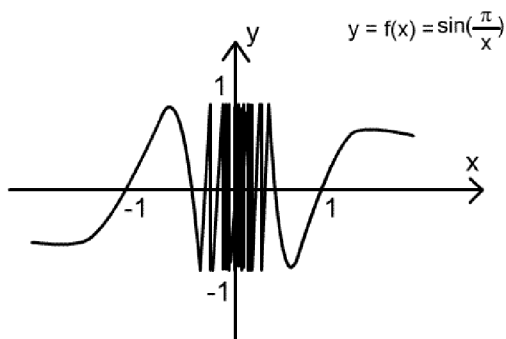
86) Define right hand rule and left hand rule?

"The limit of $f(x)$ as x approaches 0 from the right", the plus on the 0 stands for "from the right" This is called the RIGHT HAND LIMIT.

$$\lim_{x \rightarrow 0^+} \frac{\sin(x)}{x}$$

"The limit of $f(x)$ as x approaches 0 from the left", the minus on the 0 stands for "from the left" This is called the LEFT HAND LIMIT.

$$\lim_{x \rightarrow 0^-} \frac{\sin(x)}{x}$$



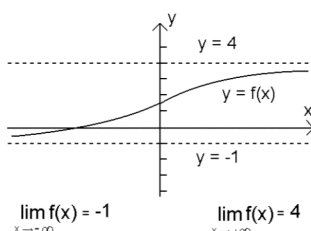
87) This Graph has NO LIMITING value as it OSCILLATES btw

- a) -1 and -1
- b) -1 and 1
- c) Both a and b
- d) 1 and -1

88) Limits fail for many reason, but usual culprits are

- a) Oscillations
- b) unbounded Increase
- c) unbounded decrease
- d) Both b and c

89) the values of $f(x) = y$ DECREASE without bound as $0 \rightarrow x$ not from both the left and the right .



90) Basic function

- a) $g(x) = x$

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b) $f(x) = k$

c) $\frac{\sin(x)}{x}$

d) Both a and b

91) $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$ is Equals to

a) 1

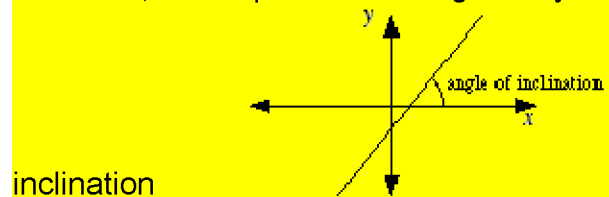
b) 2

c) 3

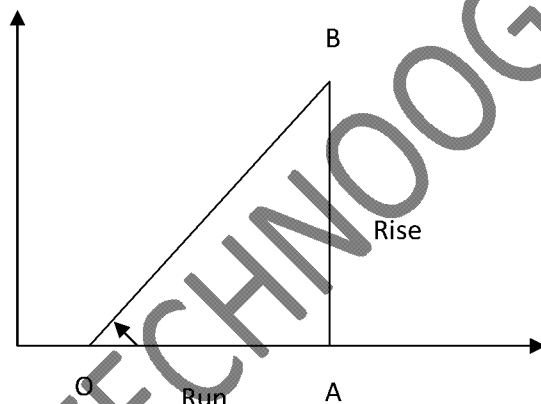
d) 0

92) What is Angle of Inclination of a Line?

Ans. The angle between a line and the x-axis. This angle is always between 0° and 180° , and is measured counter clock wise from the part of the x-axis to the right of the line. Note: All horizontal lines have angle of inclination 0° . All vertical lines have angle of inclination 90° . Also, the slope of a line is given by the tangent of the angle of



Theorem 1.4.3



In the above figure, if we take slope of the line, then it will become $m =$

$\frac{AB}{OA} \dots \dots \dots (i)$

In the above figure, you can see that if we take $\tan \theta$, then it will become

$\tan \Phi = \frac{AB}{OA} \dots \dots \dots (ii)$

From (i) and (ii), we can equate the equations, so we have $m = \tan \Phi$

93) What is absolute value?

Ans. Absolute value is the magnitude of a quantity. Suppose you and your friend walking in opposite direction from the same location (say point zero), after some time both of you cover the distance of five yards. Now if the values are to be assigned to the new locations of both of you, it is suitable to say one is at +5 and the other is at -5, but the actual value of the distance covered by each of you is 5, this is

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called the absolute value whereas the other values are shown with the negative and positive signs, these signs are used to show the direction to which they are located from the starting point. As another example if you go from Lahore to Islamabad and cover the distance of x km (kilometers), your location with respect to Lahore is x km but if you come back it will be $-x$ and you'll be in Lahore again, total distance covered by you is $2x$ which is absolute value but if it is not considered, you followed x km first and $-x$ km after this.

What are three cases for this formula, depending on value of k .

94) In the equation of circle,

$$(x - x_0)^2 + (y - y_0)^2 = k \quad \text{---(A)}$$

Ans.

There are three cases for this formula, depending on value of k .

Case (1)

When $k > 0$, the equation (A) shows a circle with radius k .

Case (2)

When $k = 0$, the equation (A) shows a circle with radius 0 which is a single point $x = x_0, y = y_0$.

Case (3)

When $k < 0$, the equation (A) has no real solution, so it shows no graph.

The cases (2) and (3) are the degenerated Case, i.e. these two cases deviate from the original form.

15) average velocity of a body is

$$\frac{d_1 - d_0}{t_1 - t_0}$$

a)

$$\frac{t_1 - t_0}{f(t_0) - f(t_1)}$$

b)

$$\lim_{t_1 \rightarrow t_0} \frac{f(t_0) - f(t_1)}{t_1 - t_0}$$

c)

d) None of these

95) Consider two function $f(x) = x^3$ and $g(x) = (x+9)$ then $f \circ g(x) =$

a)

$$(x+9)^3$$

b)

$$x+3$$

c)

$$x+9$$

d) None of these

96) Consider two function $f(x) = x^2$ and $g(x) = \sqrt{x}$ then $f \circ g(x) =$

a)

$$x^2$$

b)

$$\sqrt{x}$$

c)

$$x \text{ -correct}$$

d) None of these

97) Consider two function $f(x) = 3\sqrt{x}$ and $g(x) = \sqrt{x}$ what is true about these function

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$$\frac{f(x)}{g(x)} = 3x$$

- a)
- b) $f(g(x)) = 3x$
- c) $f(x) \cdot g(x) = 3x$
- d) None of these -correct

98) The centre and the radius of the circle $(x+5)^2 + (y-3)^2 = 16$ is

- a) (5,-3),16
- b) (5,-3),4
- c) (-5,3),4
- d) None of these

99) The graph $x = y^2$ is symmetric about

- a) Y-axis
- b) Origin
- c) X-axis
- d) None of these

100) The chain rule is used for two function f and g, if we have ----- of these function

- a) Product
- b) Sum
- c) Composition
- d) None of these

101) A function f is differentiable function if it is differentiable on the interval

- a) $(-\infty, \infty)$
- b) (a, ∞) where a is any negative integer
- c) $(0, \infty)$
- d) None of these

102) A function is said to be continuous function if the function is continuous on the interval

- a) $(-\infty, +\infty)$
- b) $(0, +\infty)$
- c) $(-\infty, 0)$
- d) None of these

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

103)

- a) 1
- b) 2
- c) 0-correct
- d) 1/2

104) For any polynomial $P(x) = c_0 + c_1x + \dots + c_nx^n$ and any real number a

$$\lim_{x \rightarrow a} P(x) = c_0 + c_1a + \dots + c_na^n =$$

- a) $P(a)$ -correct
- b) $P(a+1)$

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- c) $P(a-1)$
d) $\frac{1}{P(a)}$
- 105) no of x and y intercepts for the equation $y=1/x$
a) Two x intercepts
b) Two y intercepts
c) No x and no y intercepts-
d) None of these
- 106) A line is called a tangent line to the circle if it meets the circle at precisely
a) One point
b) Two points
c) Infinite points
d) None of these
- 107) If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f is relative At x_0
a) Minima
b) Maxima
c) None of these
d) Both a and b
- 108) If the $\lim_{x \rightarrow a} f(x) = L$ then the inequality $(L - \varepsilon) < f(x) < L + \varepsilon$ holds in any subset of the interval
a) $(a - \delta, a) \cup (a, a + \delta)$
b) $(a - 1, a) \cup (a, a + 1)$
c) $(a - \varepsilon, a) \cup (a, a + \varepsilon)$
d) None of these
- 109) The set $\{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is known as set of
a) Natural numbers
b) Integers
c) Whole numbers
d) None of these
- 110) The domain of the function $h(x) = \frac{1}{(x-2)(x-4)}$ is.
a) $(-\infty, 2) \cup (2, 4) \cup (4, +\infty)$.
b) $(-\infty, 2] \cup \{2, 4\} \cup [4, \infty)$
c) $(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$
d) All of these are incorrect
- 111) π is called
a) An integer
b) A rational number
c) An irrational number
d) A natural number
- 112) For a function $f(x)$ to be continuous on interval (a, b) the function must be continuous
a) At all point in (a, b)

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- b) Only at a and b
- c) At a mid point of a and b
- d) None of these

113) If a function satisfies the conditions

$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = f(c)$$

f(c) is defined, Exists Then the function is said to be

- a) Continuous at c
- b) Continuous from left at c
- c) Continuous from right at c
- d) None of these

$$\frac{d}{dx} [\sec x] = \text{-----}$$

114)

$$\frac{\sin x}{1 - \sin^2 x}$$

- a) $\frac{-\sin x}{1 - \sin^2 x}$
- b) $\frac{1}{1 - \sin^2 x}$
- c)
- d) None of these

$$\log_b ac = \text{-----}$$

115)

$$\log_b a + \log_b c$$

- a) $\log_a b + \log_c b$
- b) $\log_{a+c} b$
- c)
- d) None of these

$$\log_b a^r = \text{-----}$$

116)

$$a \log_b r$$

- a) $r \log_b a$
- b) $b \log_a r$
- c)
- d) None of these

$$f''(x) < 0$$

117) If on an open interval (a,b) then f is ----- on (a,b)

- a) None of these
- b) Concave up
- c) Concave down
- d) Closed

118) If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$

then f has relative At x_0

- a) Minima
- b) Maxima
- c) None of these

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$$\frac{f(x)}{g(x)} = 3x$$

- b) $f(g(x)) = 3x$
 c)
 d) None of these

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- 126) The formula _____ is called with respect to x of the function f

- a) **Derivative**
 b) Slope
 c) Tangent
 d) None of these

$$\frac{d}{dx} \left(\frac{f}{g} \right)$$

- 127) Suppose that f and g are differentiable function of x then

a) $\frac{g \cdot f' - f \cdot g'}{g^2}$

$\frac{g \cdot f' + f \cdot g'}{g^2}$

b) $\frac{g \cdot f' - f \cdot g'}{g}$

- c)
 d) None of these

- 128) $x^2 - 9 = \dots\dots\dots$

- a) $|(x-3)^2|$
 b) $|(x+3)^2|$
 c) **$|x-3||x+3|$**
 d) $|x+3||x+3|$

- 129) Usually the number that signifies the idea of $f(x)$ being as close to limit L as want to be must be a/an

- a) **Integer**
 b) Natural number
 c) Small positive number.
 d) Small negative number.

- 130) A function f is said to be continuous on a closed interval $[a, b]$ if f is discontinuous from the right at "a" and "f" is continuous from the left at "b" and "f" is continuous on

- a) $(a, b]$
 b) $[a, b)$
 c) $[a, b]$
 d) **(a, b)**

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- 131) If f is continuous on $[a, b]$, and if $f(\text{angel})$ and $f(\text{beer})$ have opposite signs, then there is one solution of the equation $f(x) = 0$ in the interval (a, b) .
- a) at most
 - b) exactly
 - c) at least**
 - d) not more than
- 132) (epsilon) used in the definition of limit can be a negative number.
- a) True
 - b) False**
- 133) If a function is differentiable at a point then it is continuous at that point. The converse is
- a) False
 - b) True**
- 134) If the function f and g are continuous at c , then $f + g$ is at
- a) Discontinuous
 - b) Continuous**
- 135) If f is continuous on a closed interval $[a, b]$ and C is any number between $f(\text{angel})$ and $f(\text{beer})$, inclusive, then there is at least one number x in the interval $[a, b]$ such that -----
- a) $f(x)$ is not equal to C
 - b) $f(x) = C$**
 - c) $f(x) > C$
 - d) $f(x) < C$
- 136) $|x-3| < 1$ implies.....
- a) $-4 < x < 4$
 - b) $2 < x < 4$**
 - c) $-2 < x < -4$
 - d) $x-3 < 1$
- 137) If for any positive number ϵ (epsilon) we can find δ (delta) such that $|(3x-5)-1| < \epsilon$, if x satisfies $0 < |x-2| < \delta$ Then $f(x) = \dots\dots\dots$
- a) $3x-5-1$
 - b) $x-2$
 - c) $3x-5$**
 - d) None of these
- 138) graph of an equation is the points on the xy - plane that the equation.
Select correct option:
- a) Satisfy**
 - b) Does not satisfy
- 139) If two lines have same slope, say 1, then these two lines are _____
Select correct option:
- a) Parallel to each other**
 - b) Parallel to x -axis
 - c) Parallel to y -axis
 - d) Perpendicular to each other
- 140) There is one-to-one correspondence between the points on co-ordinate line and _____
- a) Set of natural numbers

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- b) Set of integers
c) Set of irrational numbers
d) **Set of real numbers**
- 141) Graph of the equation $x^2+y^2=9$ represents a.....
Select correct option:
a) **Circle not sure**
b) Parabola
c) Ellipse
d) None of these.
- 142) If $\sin(3x^2) / 6 + C$ is the anti-derivative of a function $f(x)$, then $f(x) =$ _____.
a) $x^2 \cos(3x^2)$.
b) **$x \cos(3x^2)$.**
c) $x \cos(3x)$.
d) none
- 143) Which of the following is the integral of $\sin(3x+5)$ with respect to x ?
a) $-1/3[-\cos(3x+5)]$
b) **$1/3[-\cos(3x+5)]$**
c) $1/15[-\cos(3x+5)]$
d) $-\cos(3x+5)$
- 144) If 'n' goes from 1 to 3 and the summation of $'na' = 6a$, then the value of 'a' is -----
a) **6**
b) -6
c) 1
d) undetermined
- 145) If 'n' goes from 1 to any large ODD number then the summation of $'(-1)^n'$ = -----
a) **-1**
b) 0
c) 1
d) that specific large ODD number
- 146) $1+2+3+\dots+t$ equals
a) $n(n+1)/2$
b) **$t(t+1)/2$**
c) $n(n+1)(2n+1)/6$
d) none
- 147) If definite integral of $f(x)=\sin x$ over $[a,0]$ is equal to $'-2'$ then the value 'a' is-----
a) $\pi/2$
b) **π**
c) 0
d) $-\pi$
- 148) If the definite integral of $f(x)=3$ over $[1,x]$ is greater than $'12'$ then ----
a) $x>12$
b) $x>5$
c) **$x>3$**

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- d) $x > 1$
- 149) If $[-8, 8]$ is subdivided into '16' equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be-----.
- a) 2
b) 3
c) 4
d) 5
- 150) Which of the following is the integral of $\sin(2x)$?
- a) $\cos(2x) + C$
b) $2\cos(2x) + C$
c) $-(1/2)\cos(2x) + C$
d) none
- 151) Sum of cubes of n-terms of a series whose nth term is 'n' = ---
- a) Square of $n(n+1)(2n+1)/6$
b) Square of $n(n+1)/2$
c) Square of $(n+1)/2$ (just a guess)
d) Square of $n(n+1)/6$
- 152) In a rectangular coordinate system the coordinate axes divide the plane into four regions called...
- a) Cube
b) Quadrants
c) Circle
- 153) An equation of the form is called
- a) Parabolic
b) Cubic
c) Quadratic
- 154) Graph of $f(x) = |x|$ is
- a) Circle shape
b) V shaped
c) U shaped
- 155) $Y = f(x) + 3$ translation of the function is at 3 is By c units
- a) Up
b) Down
c) Left
d) Right
- 156) Graph of $y = |x - 5|$ of the graph $y = |x|$ 5 units to the right to get the graph of y.
- a) Reading
b) Translation
c) Creating
d) Definition
- 157) By the definition and be on the interval (x_0, x_1)
- a) Greater than zero
b) Equal to zero
c) Less than zero
d) Less than and equal to zero
- 158) $\cot(x)$ discontinuous at
- a) $\cos(x) = 0$

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questions for mid term 2017**

- b) $\sin(x) = 0$**
c) $\tan(x) = 0$
d) $\cot(x) = 0$
- 159) If a function is differentiable then the function is also on that point
a) Discontinues
b) No exit
c) Continuous
- 160) $\tan(5-3x)$ find differentiation w.r.t. x
a) $-3\sec^2(5-3x)$
b) $3\sec^2(5-3x)$
c) $3\cos(5-3x)$
d) $3\cos^2(5-3x)$
- 161) Derivative of the will be solved by Method
a) Implicit
b) Explicit
c) Simple
- 162) Constant function has the slope
a) Positive
b) Negative
c) Zero
- 163) Decreasing function has the slope
a) Negative
b) Positive
c) Zero
- 164) The equation of line of the form is known as
a) Point-slope form
b) Two points form
c) Intercepts form
d) Slope intercept form
- 165) The set of rational number is a subset of
a) Integers
b) Natural numbers
c) Odd integers
d) Real number
- 166) If $\sin(3x^2) / 6 + C$ is the anti-derivative of a function $f(x)$, then $f(x) =$ _
a) $x^2 \cos(3x^2)$.
b) $x \cos(3x^2)$.
c) $x \cos(3x)$.
c) none
- 167) Which of the following is the integral of $\sin(3x+5)$ with respect to x ?
a) $-1/3[-\cos(3x+5)]$
b) $1/3[-\cos(3x+5)]$
c) $1/15[-\cos(3x+5)]$
d) $-\cos(3x+5)$
- 168) If 'n' goes from 1 to 3 and the summation of 'na' = 6a, then the value of 'a' is -----
a) 6
b) -6
c) 1
d) undetermined

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- 169) If 'n' goes from 1 to any large ODD number then the summation of $(-1)^n$ = -----
a) **-1**
b) 0
c) 1
d) that specific large ODD number
- 170) $1+2+3+\dots+t$ equals
a) **$n(n+1)/2$**
b) $t(t+1)/2$
c) $n(n+1)(2n+1)/6$
d) none
- 171) If definite integral of $f(x)=\sin x$ over $[a,0]$ is equal to '-2' then the value 'a' is-----
a) $\pi/2$
b) **π**
c) 0
d) $-\pi$
- 172) If the definite integral of $f(x)=3$ over $[1,x]$ is greater than '12' then -----
a) $x>12$
b) $x>5$
c) **$x>3$ (not sure)**
d) $x>1$
- 173) If $[-8,8]$ is subdivided into '16' equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be-----
a) 2
b) 3
c) 4
c) **5**
- 174) Which of the following is the integral of $\sin(2x)$?
a) $\cos(2x)+C$
b) $2\cos(2x)+C$
c) **$-(1/2)\cos(2x)+C$**
d) none
- 175) Sum of cubes of n-terms of a series whose nth term is 'n' = ---
a) Square of $n(n+1)(2n+1)/6$
b) Square of $n(n+1)/2$
c) **Square of $(n+1)/2$ (just a guess)**
d) Square of $n(n+1)/6$
- 176) In a rectangular coordinate system the coordinate axes divide the plane into four regions called...
a) Cube
b) **Quadrants**
c) Circle
- 177) An equation of the form $y = ax^2 + bx + c$ is called
a) Parabolic
b) Cubic
c) **Quadratic**
- 178) Graph of $f(x) = |x|$ is
a) Circle shape
b) **V shaped**
c) U shaped
- 179) $Y=f(x) + 3$ translation of the function is at 3 is By c units
a) **Up**
b) Down
c) Left

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- d) Right
- 180) Graph of $y=|x-5|$ of the graph $y=|x|$ 5 units to the right to get the graph of y .
- a) Reading
- b) Translation**
- c) Creating
- d) Definition
- 181) By the definition ε and δ be on the interval (x_0, x_1)
- a) Greater than zero**
- b) Equal to zero
- c) Less than zero
- d) Less than and equal to zero
- 182) Let $f(x)$ is the function as x approaches to 0 then the limit from the positive will be ...
- $\lim_{x \rightarrow +0} x$
 $\lim_{x \rightarrow -0} x$
 $\lim_{x \rightarrow 0} x$
- 183) $\cot(x)$ discontinuous at
- a) $\cos(x) = 0$
- b) $\sin(x) = 0$**
- c) $\tan(x) = 0$
- d) $\cot(x) = 0$
- 184) If a function is differentiable then the function is also on that point
- a) Discontinues
- b) No exit
- c) Continuous**
- 185) $\tan(5-3x)$ find differentiation w.r.t. x
- a) $-3\sec^2(5-3x)$**
- b) $3\sec^2(5-3x)$
- c) $3\cos(5-3x)$
- d) $3\cos^2(5-3x)$
- 186) Derivative of the $7y^4 + x^3y + x = 4$ will be solved by Method
- a) Implicit**
- b) Explicit
- c) Simple
- 187) Constant function has the slope
- a) Positive
- b) Negative
- c) Zero**
- 188) Decreasing function has the slope
- a) Negative**
- b) Positive
- c) Zero
- 189) Consider two function $f(x) = x^2$ and $g(x) = \sqrt{x}$ then $f \circ g(x) = \dots$
- a) x**
- b) x^2

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- c) $\sqrt{x}$
- d) None of these

190) A function is said to be continuous function if the function is continuous on the interval

- a) $(-\infty, +\infty)$
- b) $(0, +\infty)$
- c) $(-\infty, 0)$
- d) None of these

191) According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number, then

$$\frac{d}{dx}[x^n] =$$

- a) x^{n-1}
- b) nx^{n-1}
- c) nx^{n+1}
- d) $(n-1)x^{n+1}$

192) $30^0 =$ _____

- a) $\frac{\pi}{3}$
- b) $\frac{\pi}{4}$
- c) $\frac{\pi}{6}$
- d) $\frac{\pi}{2}$

193) The equation of line of the form $y - y_1 = m(x - x_1)$ is known as

- a) Point-slope form
- b) two points form
- c) Intercepts form
- d) Slope intercept form

194) The set of rational number is a subset of

- a) Integers
- b) Natural numbers
- c) Odd integers
- d) Real number

195) On the straight line, the tangent at any point coincide with line -----

Select correct option:

- a) no where
- b) every where
- c) some where
- d) none of these

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Page 27

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196) On the straight line, the tangent at any point coincide with line -----

Select correct option:

- a) no where
- b) every where
- c) some where
- d) none of these**

BS.BIOTECHNOLOGY-3rd _semester